

PROJECT REPORT

Sales Management System

Branch-Based Sales Management System

Developed using Python · Streamlit · SQL Server

2026

1. Project Overview

- Sales Management System is a branch-based sales management system.
- The project helps organizations manage sales, payments, and customer records.
- It provides real-time tracking of received and pending payments.
- The system is developed using Python, Streamlit, and SQL Server.
- Role-based access is used for better security and branch management.

2. Main Objectives

- Store branch-wise sales data in a centralized database.
- Track customer payments and pending amounts automatically.
- Support installment and partial payments.
- Provide dashboards with sales reports and charts.
- Restrict access based on user roles (Admin / Super Admin).
- Run SQL analytical queries for business insights.

3. Technology Stack

Technology	Component	Purpose
Python 3	Core Language	Application logic, data processing, DB connectivity
Streamlit	Web Framework	Interactive dashboard UI and session management
SQL Server Express	Database	Relational data storage, queries, constraints
pyodbc	DB Connector	Python to SQL Server connection via ODBC Driver 17
pandas	Data Library	Reading SQL query results into DataFrames

4. Database Design

The Sales_management_System database contains four relational tables with proper primary keys, foreign keys, and constraints. The schema ensures data integrity and supports automated financial calculations.

Table Name	Key Columns	Description
branches	branch_id (PK), branch_name, city	Stores all branch locations
users	user_id (PK), username, role, branch_id (FK)	Login credentials and role assignments
customer_sales	sale_id (PK), branch_id (FK), customer_name, gross_sales, received_amount, status	Main sales transactions table
payment_splits	payment_id (PK), sale_id (FK), amount_paid, payment_method, payment_date	Installment and partial payments

5. Application Modules & Screenshots

The Streamlit application is structured into six navigation modules. Role-based filtering is applied throughout — Super Admin accesses all branch data while Admin users are restricted to their assigned branch.

5.1 Login Page

Users authenticate with a username and password. On successful login, session state stores the username, role, and assigned branch. The login page uses a centered card layout with a gradient background.

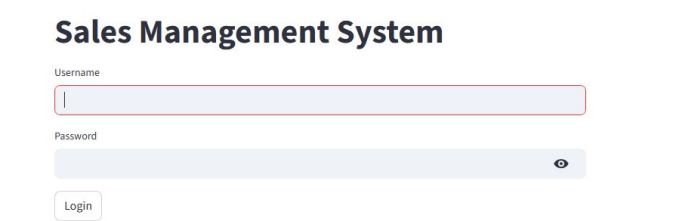


Figure 1: Login Page — Username & Password Authentication

5.2 Sales Dashboard (Overview)

Displays four KPI cards: Total Sales, Received Amount, Pending Amount, and Collection %. Dynamically filtered by Branch, Product, Start Date, and End Date. Super Admin sees all branches; Admin sees only their branch.

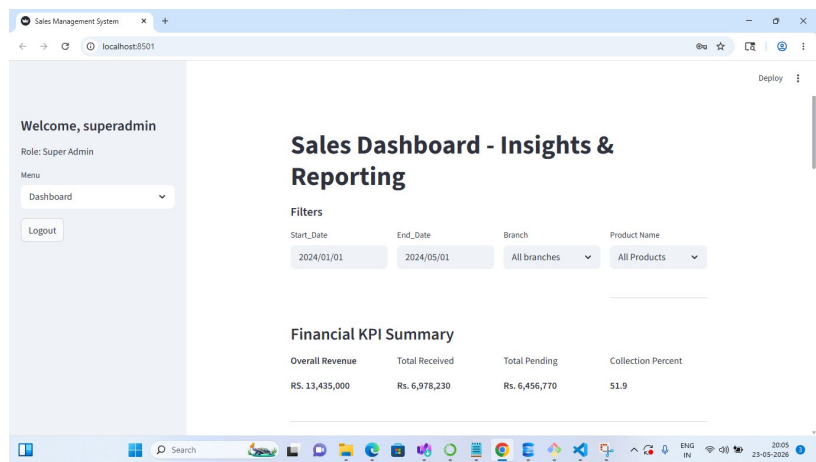


Figure 2: Sales Dashboard — KPI Cards with Branch, Product & Date Filters

5.3 Customer Sales Table

Shows all columns — branch_id, sale_date, customer name, mobile number, product_name, gross_sales, received_amount, status, sale_id, and pending_amount (gross_sales minus received_amount).

Welcome, superadmin

Role: Super Admin

Menu

Dashboard

Logout

All Sales Data

	sale_id	branch_name	date	name	product_name	gross_sales	received_s
149	851	Pune	2024-05-01 00:00:00	Customer_851	BI	28000	
150	850	Chennai	2024-04-30 00:00:00	Customer_850	DA	35000	
151	849	Chennai	2024-04-29 00:00:00	Customer_849	DS	40000	
152	848	Delhi	2024-04-28 00:00:00	Customer_848	SQL	25000	
153	847	Pune	2024-04-27 00:00:00	Customer_847	DA	35000	
154	846	Pune	2024-04-26 00:00:00	Customer_846	AI	48000	
155	845	Bangalore	2024-04-25 00:00:00	Customer_845	SQL	25000	
156	844	Bangalore	2024-04-24 00:00:00	Customer_844	DA	35000	
157	843	Ahmedabad	2024-04-23 00:00:00	Customer_843	ML	42000	
158	842	Bangalore	2024-04-22 00:00:00	Customer_842	SQL	25000	

Figure 3: Customer Sales Table — All Transactions with Status & Pending Amount

5.4 Sales Charts

Two bar charts are rendered: Branch Wise Sales (8 city branches) and Product Wise Sales (DS, DA, BA, FSD, ML, AI, BI, SQL).

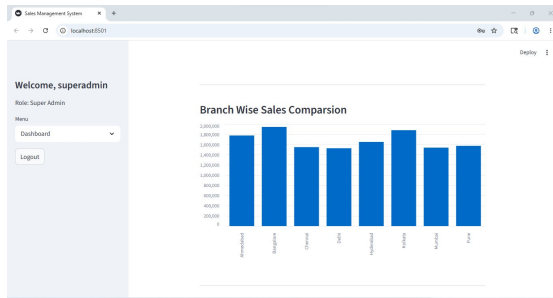


Figure 4: Branch Wise Sales — Bar Chart

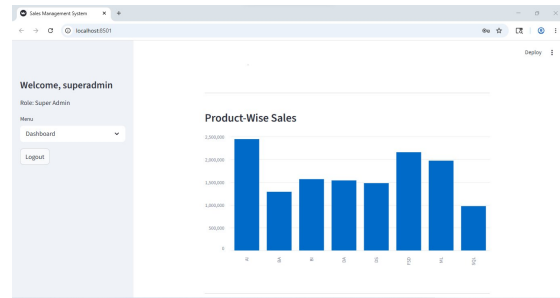


Figure 5: Product Wise Sales — Bar Chart

5.5 Sales Entry

A form to add a new customer sale. Fields: Customer Name, Mobile Number, Customer Join Date, Product (DS, DA, BA, FSD, AI, ML, BI, SQL), Gross Sales, and Branch. received_amount defaults to 0 and status defaults to 'Open' on insert.

The 'Add New Sale' form in the Sales Management System. It includes a sidebar with 'Welcome, superadmin' and a menu with 'Add Sale' and 'Logout'. The main form fields are: Branch (dropdown), Customer Name, Mobile number, Course Name (AI, ML, SQL, DS, DA, BA, FSD, BI), Gross_Sales / Course Fee (0.00), Sale Date (2026/05/23), and an 'Add Sale' button.

Figure 6: Sales Entry Form — Add New Customer Sale

5.6 Payment Splits

Handles partial and installment-based payments. User enters Sale ID, Payment Date, Amount Paid, and Payment Method (Cash / UPI / Card). Auto-closes sale when received_amount $\geq$ gross_sales.

The 'Add Payment' form in the Sales Management System. It includes a sidebar with 'Welcome, superadmin' and a menu with 'Add Payment' and 'Logout'. The main form fields are: Select Sale (Customer_1000 - Pending Rs. 15353.0), Amount Paid (0.00), payment_method (Cash), and an 'Add Payment' button.

Figure 7: Payment Splits — Add Installment Payment for a Sale

6. Results & Conclusion

6.1 Results Achieved

- Fully functional multi-branch Sales Management System built and deployed.
- SQL Server database with 4 normalized tables, foreign key relationships, and constraints.
- Automated payment tracking — received amount and status update on every payment entry.
- Role-based access working correctly — Super Admin views all branches, Admin restricted to own branch.
- Interactive Streamlit dashboard with KPI cards, filterable sales table, and 2 bar charts.
- All 20 SQL analytical queries implemented and executable from the dashboard with CSV export.
- Payment splits support partial and installment payments with full payment history view.

6.2 Conclusion

The Sales Management System successfully delivers a complete, production-ready branch-based sales management solution. It combines a well-normalized SQL Server database with an automated payment tracking mechanism and a clean Streamlit web interface. The system provides real-time financial visibility through KPI cards and charts, enforces role-based data security, and supports all 20 required SQL analytical queries — demonstrating strong command of relational database design, Python application development, and data-driven dashboard building.